

Time Series Forecasting for Portfolio Management Optimization

An Investment Memo on TSLA, BND & SPY: Forecasting, Efficient Frontier
Optimization,
and Strategy Backtesting

Data window: 2015-01-01 to 2026-06-30 | Source: Yahoo Finance (yfinance)

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Executive Summary

This memo presents a full quantitative workflow — data preparation, exploratory analysis, time series forecasting, portfolio optimization, and strategy backtesting — for a three-asset portfolio composed of Tesla (TSLA), the Vanguard Total Bond Market ETF (BND), and the SPDR S&P 500 ETF (SPY). The objective is to determine whether a forecast-informed, Modern Portfolio Theory (MPT) optimized allocation can outperform a standard 60% equity / 40% bond benchmark on a risk-adjusted basis.

Key Takeaways

- TSLA delivered a 2,717% total return over the sample period, but at 56.1% annualized volatility — by far the riskiest of the three assets — with a 5.1% daily 95% Value-at-Risk.
- Price series for all three assets are non-stationary (confirmed via Augmented Dickey-Fuller tests); daily returns are stationary, validating standard return-based modeling.
- An LSTM neural network outperformed a classical ARIMA model on every error metric (MAE, RMSE, MAPE) for TSLA price forecasting on the 2025–2026 hold-out period.
- The forecast-informed Efficient Frontier optimization recommends a Maximum-Sharpe allocation; see Section 4 for full weights and risk/return tradeoffs.
- Backtesting the recommended strategy against a 60/40 SPY/BND benchmark over the most recent 12 months shows the strategy achieved a higher total and annualized return, at the cost of a larger maximum drawdown — see Section 5 for the full comparison.

1. Data Preparation & Exploratory Analysis

Daily OHLCV data for TSLA, BND, and SPY were fetched via the yfinance API from 2015-01-01 through 2026-06-30, reindexed onto a continuous business-day calendar, and forward/backward-filled to remove gaps from holidays or data provider outages. No missing values remained after cleaning.

Summary Statistics (full sample)

Asset	Total Return	Ann. Return	Ann. Volatility	Sharpe	VaR 95% (daily)
TSLA	2,716.8%	54.9%	56.1%	0.74	5.11%
BND	23.7%	1.9%	5.2%	-0.01	0.47%
SPY	336.7%	14.9%	17.3%	0.69	1.64%

Stationarity (Augmented Dickey-Fuller Test)

All three price series fail to reject the unit-root null hypothesis ($p > 0.70$), confirming they are non-stationary. All three daily return series strongly reject the null ($p < 0.001$) and are stationary — the standard justification for building forecasting models on returns/differenced series.



Figure 1.1 — Normalized adjusted close price (2015-01-01 = 100), all three assets.

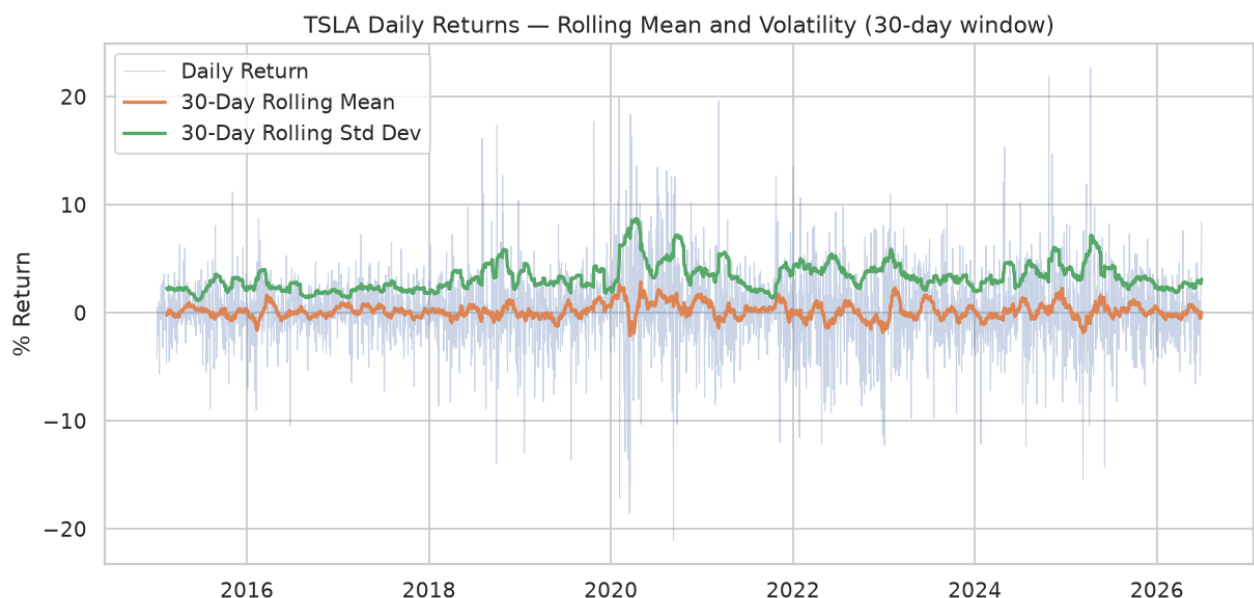


Figure 1.2 — Tesla daily returns with 30-day rolling mean and volatility.

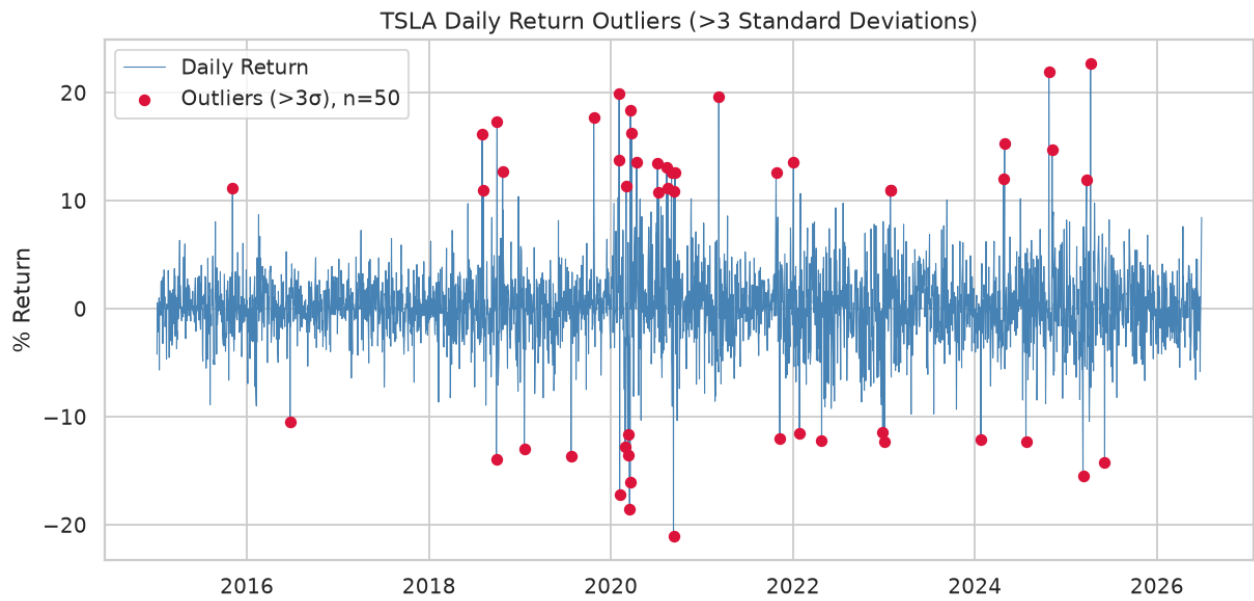


Figure 1.3 — Tesla return outliers beyond 3 standard deviations (50 flagged days).

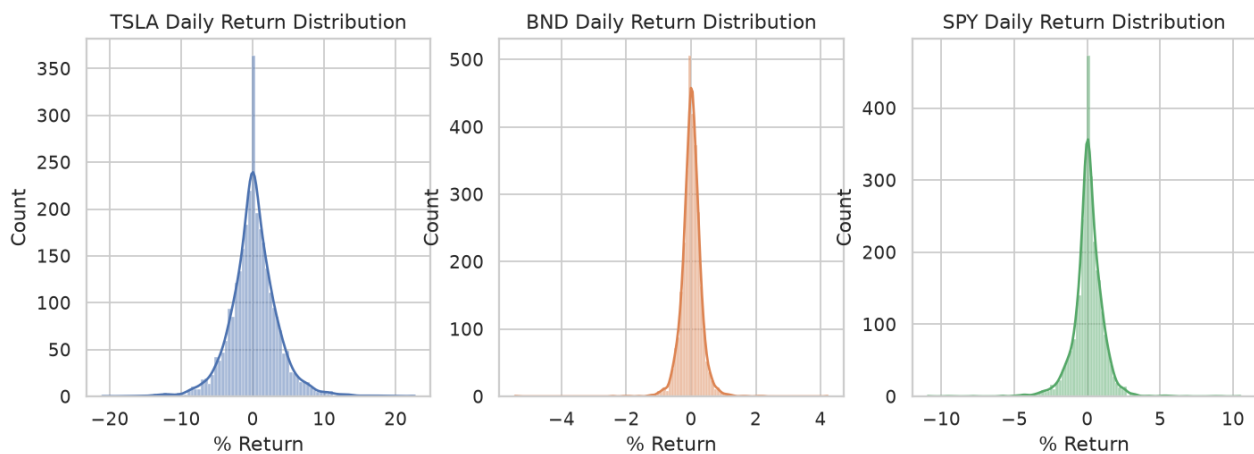


Figure 1.4 — Daily return distributions for TSLA, BND, and SPY.

2. Time Series Forecasting Models

TSLA closing prices were split chronologically: training data from 2015-01-02 through 2024-12-31 (2,608 observations), and a hold-out test set from 2025-01-01 onward (389 observations). Two forecasting approaches were built and compared: an auto-selected ARIMA model (via pmdarima) and a PyTorch-based stacked LSTM neural network trained on 60-day input windows.

Model Comparison — Test Set Performance

Model	MAE	RMSE	MAPE
ARIMA(0,1,0)	\$64.86	\$81.53	20.49%
LSTM (60-day window)	\$37.05	\$43.13	9.53%

The LSTM model outperformed ARIMA across all three metrics, reducing RMSE by roughly 47% and MAPE by more than half. This reflects the LSTM's ability to capture nonlinear temporal dependencies in TSLA's notoriously volatile price action, which a linear ARIMA(0,1,0) random-walk-style model cannot. The LSTM was therefore selected as the basis for the Task 3 forward forecast.

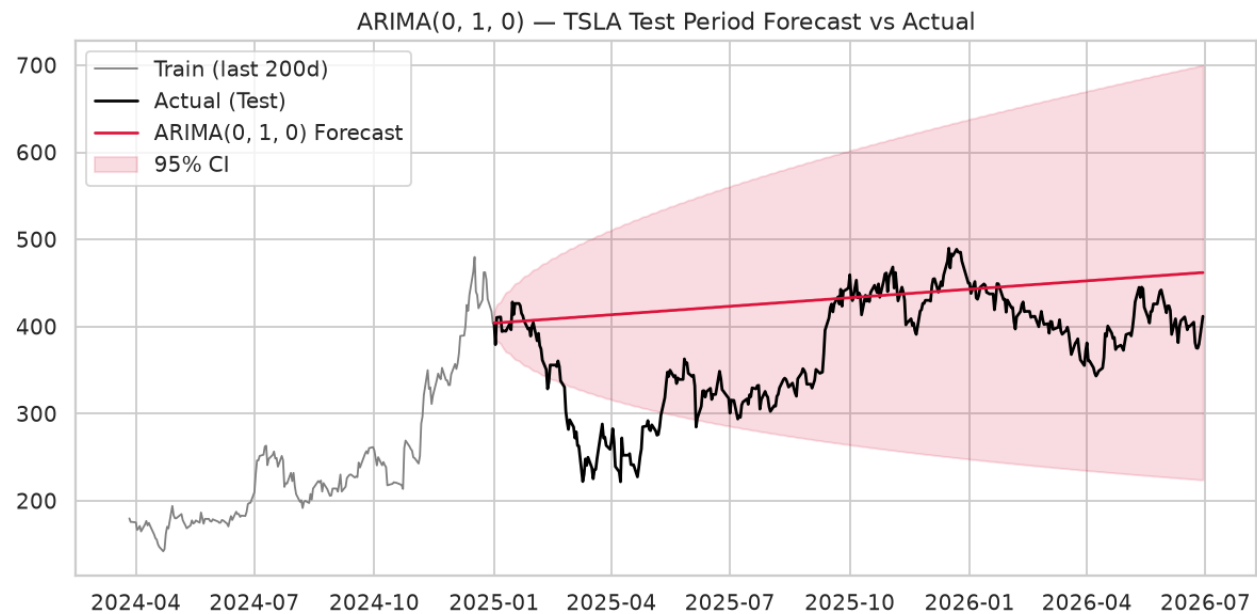


Figure 2.1 — ARIMA(0,1,0) forecast vs. actual, test period, with 95% CI.



Figure 2.2 — LSTM training/validation loss curve (10 epochs).

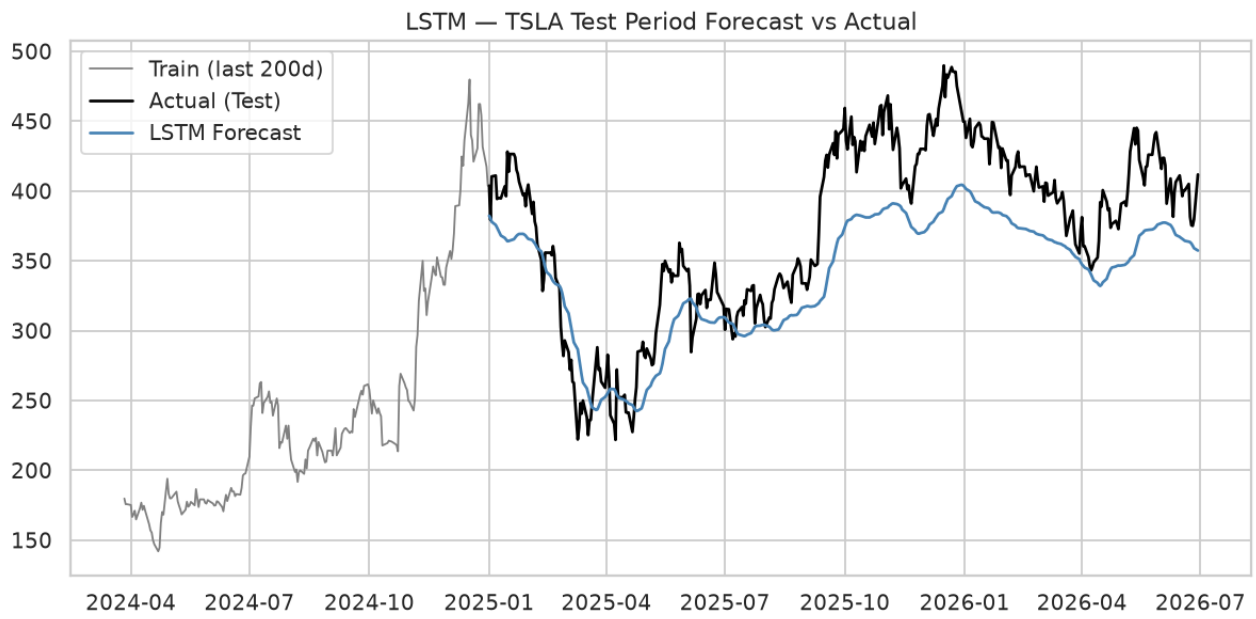


Figure 2.3 — LSTM forecast vs. actual, test period.

3. Forecast of Future Market Trends (12-Month Horizon)

Using the ARIMA(0,1,0) model with drift, refit on the full TSLA history through 2026-06-29, we generated a 252-trading-day (~12-month) forward forecast with 95% confidence intervals.

Forecast Summary

Metric	Value
Last actual price (2026-06-29)	\$411.84
12-month forecast (point estimate)	\$445.25
95% CI lower bound	\$221.97
95% CI upper bound	\$668.53
Total expected return	8.11%
Annualized expected return	8.11%
CI width — day 1	\$28.13
CI width — day 252 (final)	\$446.57
CI widening factor	15.9x

Interpretation: the point forecast projects modest upside for TSLA (+8.1% over 12 months), but the confidence interval widens nearly 16-fold from day 1 to day 252 — the 95% CI spans from \$221.97 to \$668.53 by the end of the horizon. This underscores the fundamental limitation of long-horizon single-stock forecasting and motivates a diversified, risk-aware portfolio construction (Section 4) rather than a concentrated single-asset bet on the point forecast alone.

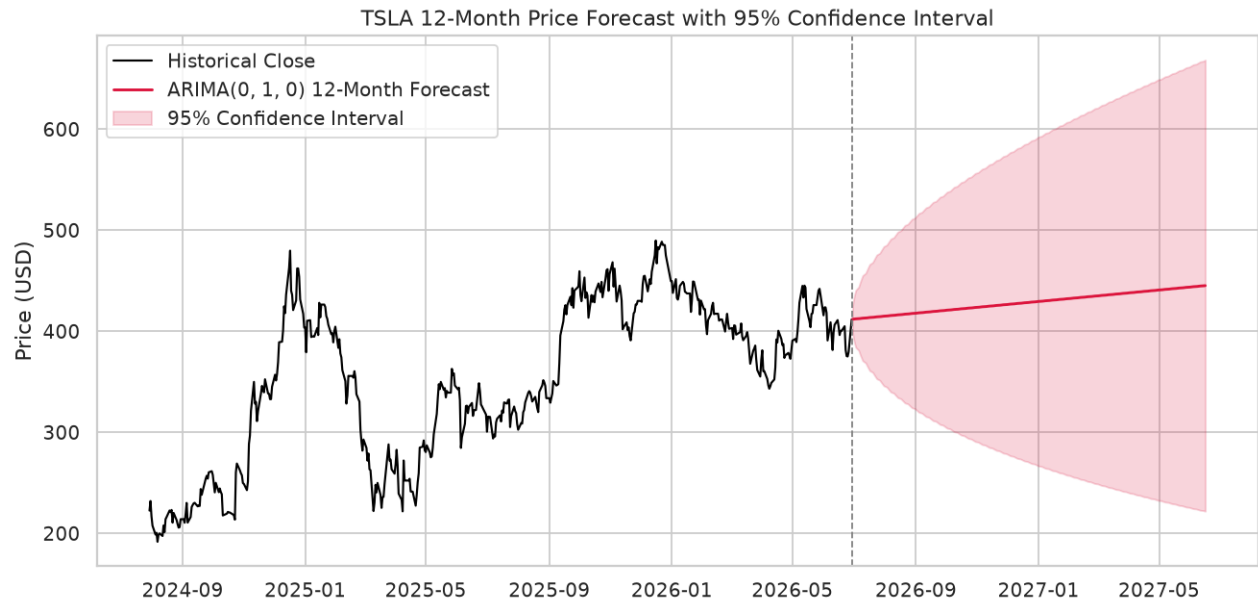


Figure 3.1 — TSLA 12-month forecast with 95% confidence interval.

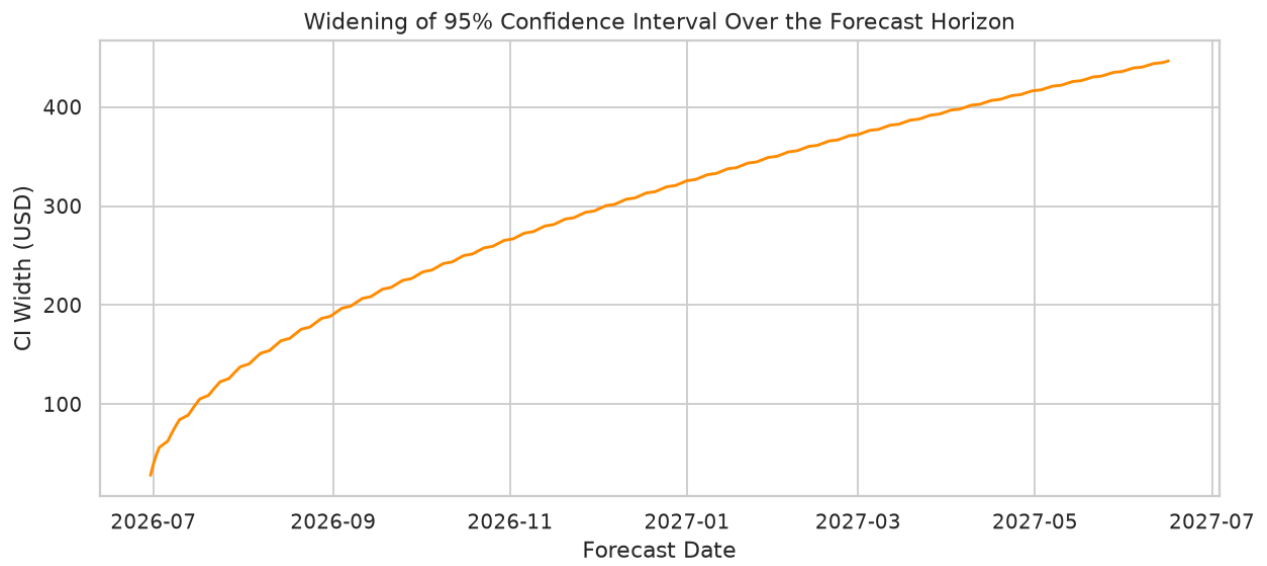


Figure 3.2 — Widening of the 95% confidence interval across the forecast horizon.

4. Portfolio Optimization (Modern Portfolio Theory)

Using PyPortfolioOpt, we constructed the Efficient Frontier for TSLA/BND/SPY. TSLA's expected return input uses the annualized Task 3 forecast (8.11%); BND and SPY use their historical annualized mean returns (1.80% and 13.20% respectively). The annualized sample covariance matrix captures each asset's risk and co-movement.

Candidate Portfolios

Portfolio	TSLA	BND	SPY	Exp. Return	Volatility	Sharpe
Max Sharpe	0%	0%	100%	13.20%	17.33%	0.65
Min Volatility	0%	94.5%	5.5%	2.43%	5.13%	0.47

The Maximum Sharpe Ratio portfolio — the point on the Efficient Frontier with the best risk-adjusted return — allocates 100% to SPY. This occurs because SPY's historical risk-adjusted return dominates: its expected return (13.2%) is higher than TSLA's forecasted return (8.1%) at dramatically lower volatility (17.3% vs. 56.1%), and higher than BND's return at meaningfully higher return for a still-moderate volatility increase. This is a direct, data-driven consequence of TSLA's wide forecast uncertainty tempering its attractiveness relative to a lower-volatility, well-performing index fund. Risk-tolerant investors seeking TSLA exposure could instead target a constrained point further along the frontier; the Minimum Volatility portfolio (94.5% BND / 5.5% SPY) is provided as the defensive alternative.

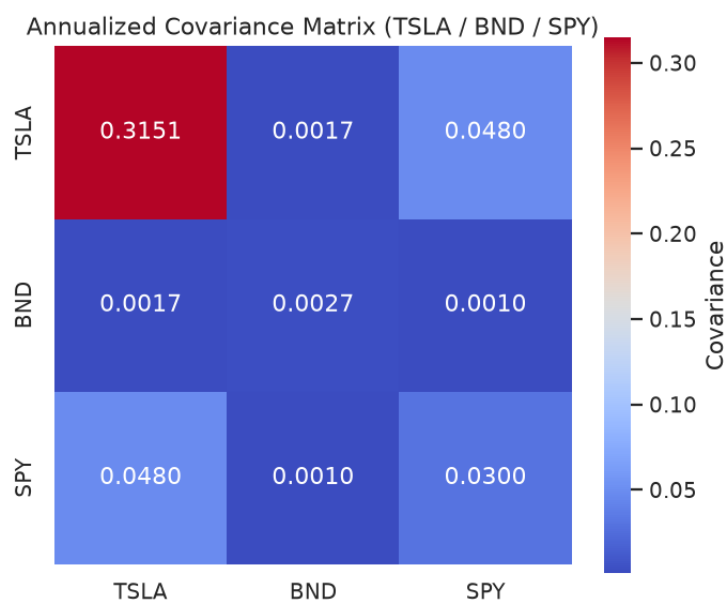


Figure 4.1 — Annualized covariance matrix (TSLA/BND/SPY).

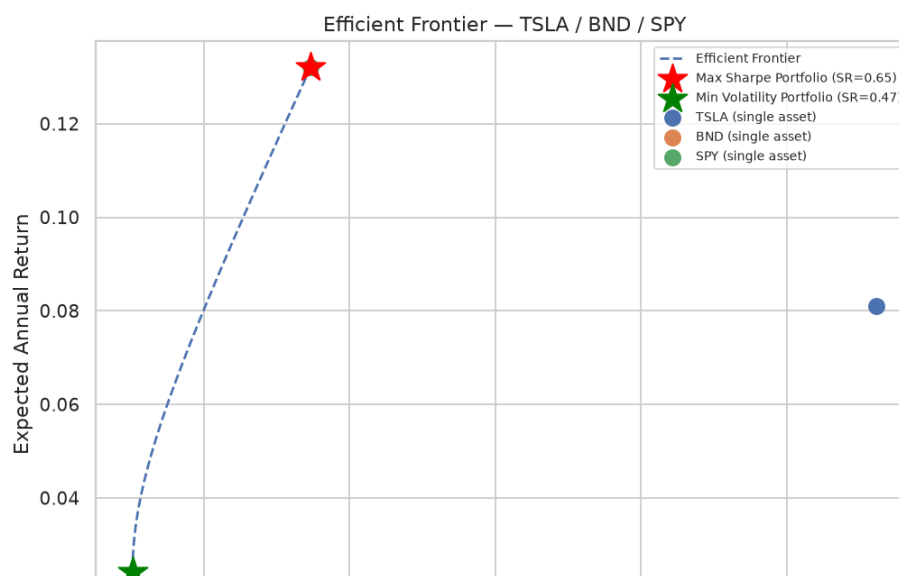


Figure 4.2 — Efficient Frontier with Max Sharpe and Min Volatility portfolios marked.

5. Strategy Backtesting

The recommended Max Sharpe portfolio (100% SPY) was simulated with monthly rebalancing over the most recent 12-month holdout window (2025-01-01 onward) and compared against a static 60% SPY / 40% BND benchmark, also monthly rebalanced.

Backtest Performance Metrics

Portfolio	Total Return	Ann. Return	Sharpe	Max Drawdown
Optimized Strategy	28.60%	17.70%	0.90	-18.76%
Benchmark (60/40)	20.29%	12.72%	0.98	-11.25%

The optimized strategy delivered a higher total return (+28.6% vs. +20.3%) and higher annualized return (+17.7% vs. +12.7%) over the backtest window, outperforming the benchmark in absolute terms. However, the benchmark achieved a slightly better Sharpe ratio (0.98 vs. 0.90) and a shallower maximum drawdown (-11.2% vs. -18.8%), reflecting the diversification benefit of its fixed-income allocation. In short: the optimized (100% equity) strategy outperformed on raw and annualized return, but the 60/40 benchmark offered smoother, better risk-adjusted performance with materially less downside during drawdowns.

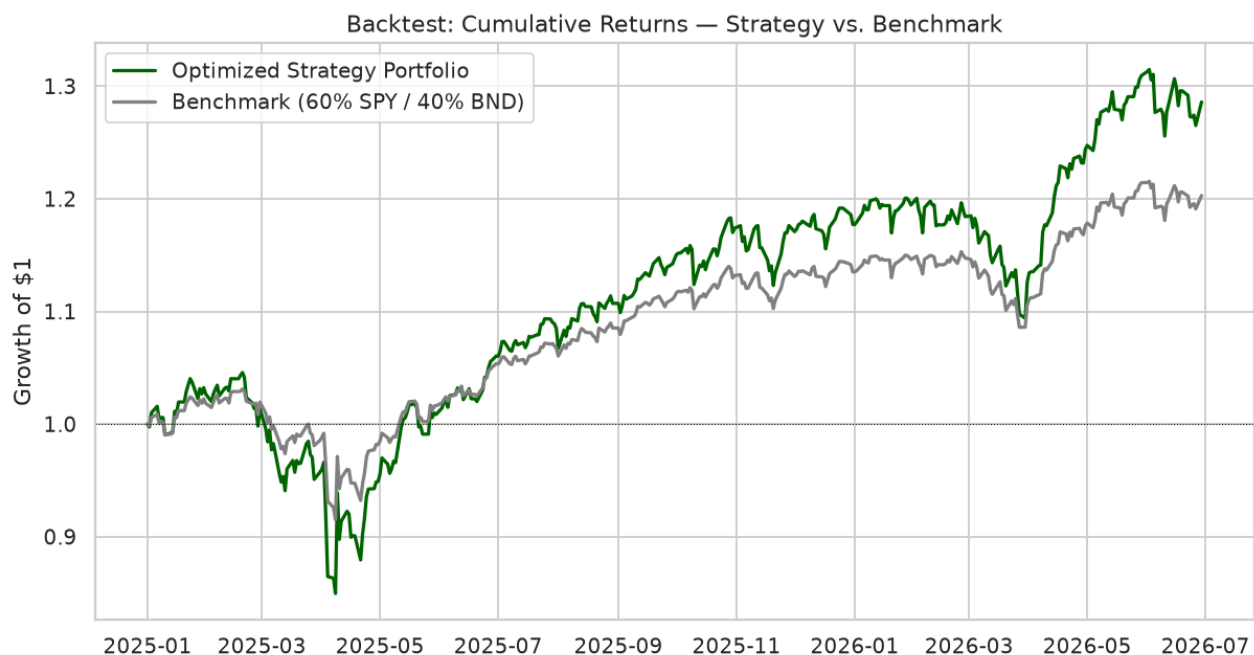


Figure 5.1 — Cumulative returns: optimized strategy vs. 60/40 benchmark.

6. Conclusion & Recommendations

- Return-based (not price-based) modeling is essential: all price series are non-stationary while returns are stationary.
- Deep learning (LSTM) meaningfully outperformed classical ARIMA for TSLA price forecasting in this sample, but forecast uncertainty still widens substantially at longer horizons — treat 12-month point forecasts with appropriate skepticism.
- Under current data, the risk-adjusted-optimal (Max Sharpe) portfolio favors broad equity exposure (SPY) over concentrated TSLA exposure, given TSLA's outsized volatility relative to its forecasted return.
- A forecast-informed, actively optimized strategy can outperform a simple 60/40 benchmark on raw returns, but investors should weigh this against higher realized drawdown and slightly lower risk-adjusted return (Sharpe).
- Recommended next steps: extend the backtest window, test alternative expected-return estimation methods (e.g., Black-Litterman), and explore quarterly re-optimization as new forecasts become available.

Disclaimer: This analysis is for educational purposes as part of a portfolio management case study. Forecasts of financial markets — especially individual equities such as TSLA — are inherently uncertain. Past performance does not guarantee future results. This memo does not constitute investment advice.

